

1. Differentiate each of the following. Simplify your answers.

$$f(x) = \frac{3x}{5} - \frac{5}{3x} - 4 + 2\sqrt[4]{x}$$

$$f(x) = \frac{3}{5}x - \frac{5}{3}x^{-1} - 4 + 2x^{\frac{1}{4}}$$

$$f'(x) = \frac{3}{5} + \frac{5}{3}x^{-2} + \frac{1}{2}x^{-\frac{3}{4}}$$

$$f'(x) = \frac{5}{3x^2} + \frac{1}{2\sqrt[4]{x^3}} + \frac{3}{5}$$

$$y = \frac{20x^5 + 8x^2 + 4x}{6x^3}$$

$$y = \frac{10x^2}{3} + \frac{4x^{-1}}{3} + \frac{2x^{-2}}{3}$$

$$y' = \frac{20}{3}x - \frac{4x^{-2}}{3} - \frac{4x^{-3}}{3}$$

$$y' = \frac{20}{3}x - \frac{4}{3x^2} - \frac{4}{3x^3}$$

$$y = 4x^{\frac{2}{3}}(x - 5\pi^3)$$

$$y = 4x^{\frac{5}{3}} - 20\pi^3 x^{\frac{2}{3}}$$

$$y' = \frac{20}{3}x^{\frac{2}{3}} - \frac{40\pi^3}{3}x^{-\frac{1}{3}}$$

$$y' = \frac{20}{3}\sqrt[3]{x^2} - \frac{40\pi^3}{3\sqrt[3]{x}}$$

$$f(x) = \frac{\sqrt{3}x^4}{3} + \frac{7}{3x^{\frac{2}{3}}} + 21\sqrt[4]{x} - 3\pi^4$$

$$f(x) = \frac{\sqrt{3}}{3}x^4 + \frac{7}{3}x^{-\frac{2}{3}} + 21x^{\frac{1}{4}} - 3\pi^4$$

$$f'(x) = \frac{4\sqrt{3}}{3}x^3 - \frac{14}{9}x^{-\frac{5}{3}} + \frac{21}{17}x^{-\frac{16}{17}}$$

$$f'(x) = \frac{4\sqrt{3}}{3}x^3 - \frac{14}{9\sqrt[3]{x^5}} + \frac{21}{17\sqrt[17]{x^{16}}}$$

$$y = 2x\sqrt{x}(x + 7\sqrt{x} - 8)$$

$$y = 2x^{\frac{5}{2}} + 14x^2 - 16x^{\frac{3}{2}}$$

$$y' = 5x^{\frac{3}{2}} + 28x - 24x^{\frac{1}{2}}$$

$$y' = 5\sqrt{x^3} - 24\sqrt{x} + 28x$$

$$g(x) = \frac{3\pi^{\frac{2}{3}}(4x^2 + 9\sqrt{x})}{x}$$

$$g(x) = 3\pi^{\frac{2}{3}}[4x + 9x^{-\frac{1}{2}}]$$

$$g'(x) = 3\pi^{\frac{2}{3}}[4 - \frac{9}{2}x^{-\frac{3}{2}}]$$

$$g'(x) = 3\pi^{\frac{2}{3}}[4 - \frac{9}{2\sqrt{x^3}}]$$

2. Determine the equation of the tangent line to the curve $y = x + \frac{1}{x^2}$ that is parallel to the line $x + y + 3 = 0$.

$$y' = 1 - \frac{2}{x^3}$$

$$1 - \frac{2}{x^3} = -1$$

$$\frac{-2}{x^3} = -2$$

$$x^3 = 1$$

$$x = 1$$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - 2 = -(x - 1)$$

$$y = -x - 3$$

$$\Rightarrow m_t = -1$$